



Data Backup and Archiving Policy

Overview

Data backup is the process of creating a copy of data to a separate medium so that it can be recovered in case the original copy is lost or a previous version is required. Data archiving is the process of removing data from immediate storage, such as a hard disk drive, and placing it on media for long term storage.

Purpose

The purpose of this document is to provide guidelines for periodic backup and archiving of data residing on servers and client PCs.

Policy

Media

- Media must have the capacity to store a full backup on a maximum of four tapes, preferably on a single tape with great capacity.
- Media must be from a named brand source with a history of quality and reliability.
- Media must be secured by means of a certified media safe.
- Backup media must be located in a building separate from the machines on which the originals reside.
- Media must be clearly labeled with an identification marking, matching that given in the backup software.

Backup Method

In order to provide long term data protection the standard TLD backup methodology employed is the Grandfather-Father-Son method. A brief description, taken from Seagate.com, is given below.

The exact period to be used by TLD MIS is to be ONE YEAR of MONTHLY backups.

Daily Tape Sets, Weekly Cycle

Monday Tuesday Wednesday Thursday W(x)

Weekly Tape Sets, 1 Month Cycle

W1 W2 W3 M(y)

Monthly Tape Sets, 1 Year Cycle

M1 M2 M3 M4 M5 M6 M7 M8 M9 M10 M11 M12

One of the most commonly used tape rotation schemes is called Grandfather-Father-Son.

This scheme uses three sets of tapes for daily, weekly and monthly backup sets. To execute the Grandfather-Father-Son rotation scheme, you will need twelve sets of media. The actual number tapes required in each media set will vary depending on how much data you need to backup.

The first set, "Son," represents your daily backups. Assign four tapes as incremental daily backups and label them "Monday" through "Thursday." These tapes will be

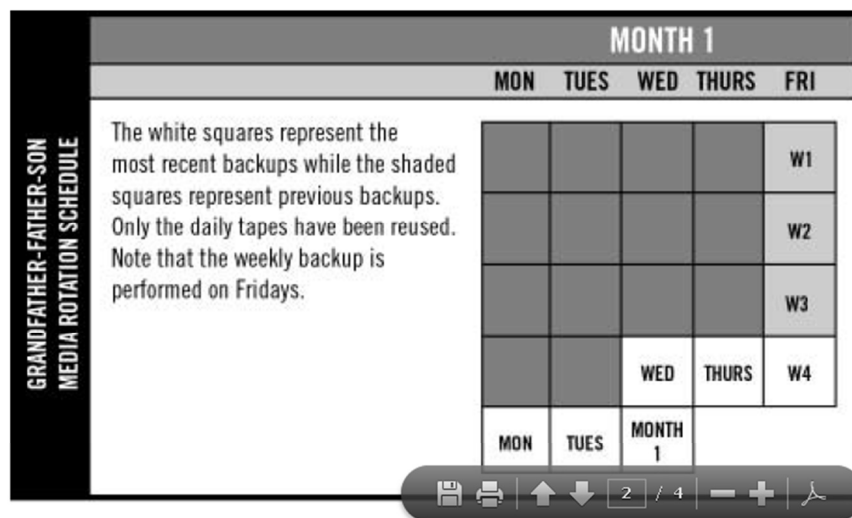
used to perform daily incremental backups and can be reused weekly on the day that they are labeled.

A second set of up to five weekly tapes, called "Father," is used to perform full backups on the day in which you do not perform a daily incremental backup. This media set should be labeled "Week 1" through "Week 5" and can be reused monthly on the day matching its label.

The final set of tapes, called "Grandfather," is used to perform full backups on the last business day of each month.

This rotation scheme will back up data on a daily, weekly, monthly. In some instances, archived data is required for periods longer.

In these cases, media sets are often pulled from the rotation and store off-site. An illustration of the Grandfather-Father-Son rotation scheme is located below.



Jobs

- Backup jobs are to be scheduled according to a Grandfather-Father-Son rotation.
- If a job was not successful it must be run at the next available opportunity.
- Backup job logs must be inspected daily to ensure previous day's job was successful.

Servers

- Mission critical data is to be backed up daily.
- At least one full backup to be done per week.
- If full backup is not done daily then daily differential backups must be performed in addition to the weekly full backup stipulated above.

Engineering Workstations

- No engineering data is to be stored on workstations.
- If data must be copied to the workstation, to be worked on locally, then it must be copied back to the server by the end of the current day.
- It is up to the user to determine whether copying a file back will disrupt others who require access to the same files.

Laptops

- Laptop data must be backed up to server storage at least once a week.
- MIS will provide tools and instructions in order to synchronize laptop data to server.
- Outlook PST files must be archived at least annually with each years' worth of emails occupying its own PST file e.g. archive2011.pst, archive2012.pst, etc.

- All data must be saved in either of the following locations in order for MIS to properly identify and back it up.
 - My Documents folder
 - Desktop
- Data outside of the above mentioned locations is the responsibility of the laptop user to backup.

Enforcement

- It is the responsibility of local IT Managers to ensure monitoring and enforcing of data backup according to the User Data Backup Audit Procedure.

User Data Backup and Archive Audit Procedure

- A weekly email is to be sent to remind the user when to perform their backup.
- The user is to confirm that the backup was done by replying to the reminder email.
- **Every year** ALL users are to be audited and backups inspected according to the above procedure.
- If a user consistently fails to perform their backup then the situation is to be brought to the attention of the user's manager and the CIO TLD Group.
 - At each end of year, require each dept. manager to take some time to move file and data that won't be modified anymore, but that cannot be simply deleted.
 - At the end of December, save that whole "archive" folder to an "archive" tape, and keep it in the safe for 5 years (maybe more or less, depending on what is being saved, and local law)
 - Then, we won't never ever save those files to tape anymore, unless department manager requires us to move those file back from the USB hard drive to the server hard drive...

Definitions

Mission Critical: data that affects the daily operation of TLD.

Full Backup: Copying of ALL data to backup media.

Differential Backup: Copying of only modified data since the last full backup.

Revision History

2006-02-27, Initial release

2012-07-20, last release, Audit part updated, logo updated